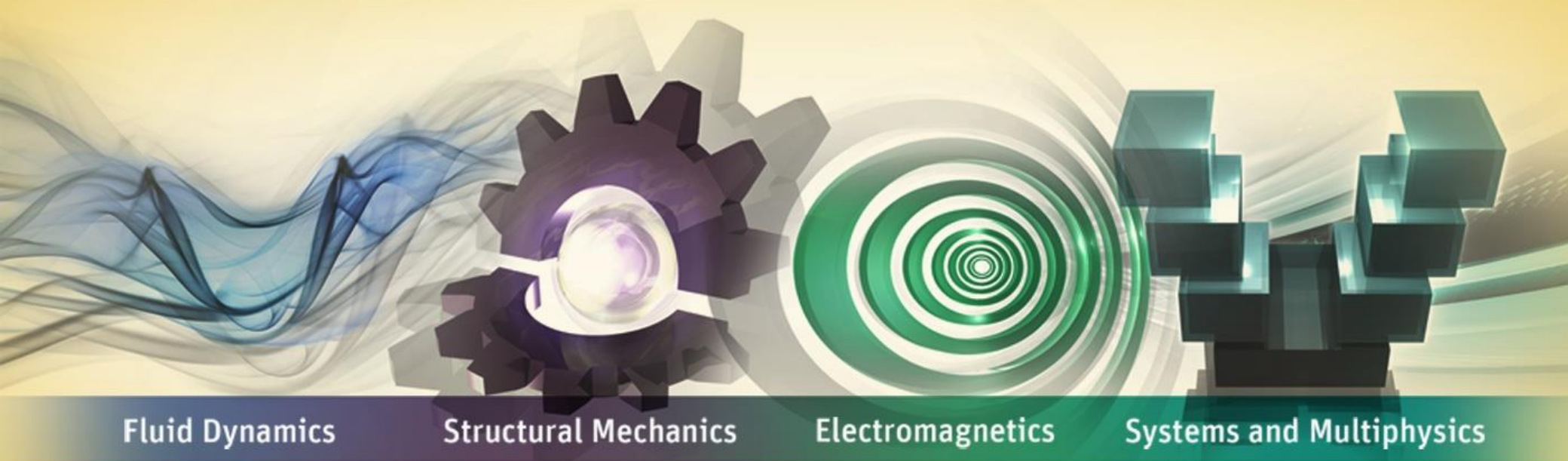


## Workshop 2B: Performing a restart



# ANSYS Mechanical Introduction to Structural Nonlinearities

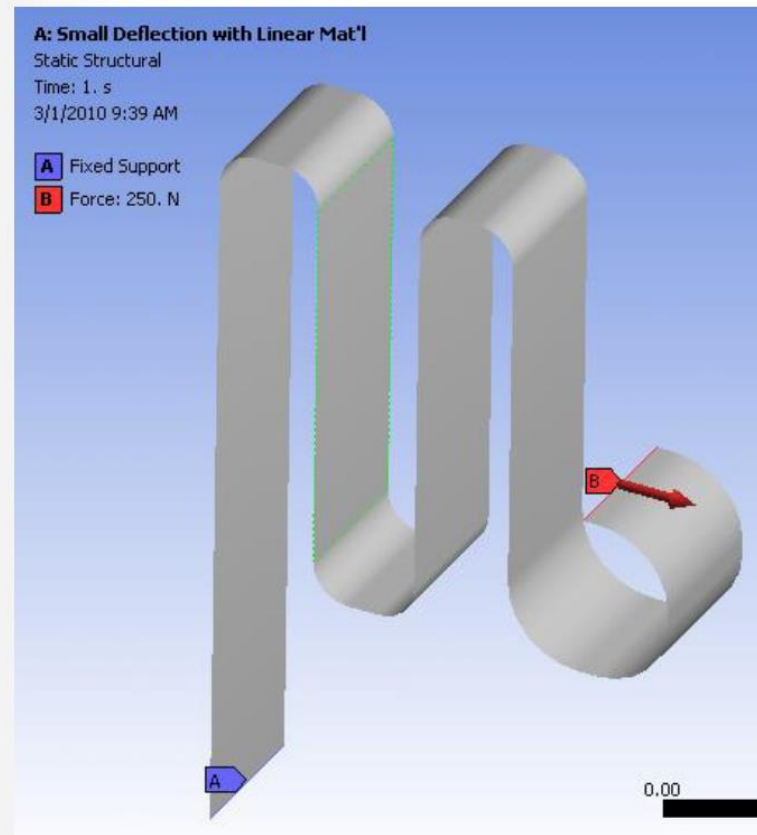
# Workshop 2B: Performing a Restart

## Goal

Use Restart Control technology to remedy a nonconverged solution and add a new load step

## Model Description

- Same as Workshop 2A



## ... Workshop 2B: Performing a Restart

Continuing with same model from Workshop 2A, highlight the Force load and increase the magnitude to 350N.

Details of "Analysis Settings"

|                                 |                    |
|---------------------------------|--------------------|
| <b>Step Controls</b>            |                    |
| Number Of Steps                 | 1,                 |
| Current Step Number             | 1,                 |
| Step End Time                   | 1, s               |
| Auto Time Stepping              | On                 |
| Define By                       | Substeps           |
| Initial Substeps                | 1,                 |
| Minimum Substeps                | 1,                 |
| Maximum Substeps                | 4,                 |
| <b>Solver Controls</b>          |                    |
| Solver Type                     | Program Controlled |
| Weak Springs                    | Program Controlled |
| Solver Pivot Checking           | Program Controlled |
| Large Deflection                | On                 |
| Inertia Relief                  | Off                |
| <b>Rotordynamics Controls</b>   |                    |
| <b>Restart Controls</b>         |                    |
| Generate Restart Points         | Manual             |
| Load Step                       | All                |
| Substep                         | All                |
| Maximum Points to Save Per Step | All                |
| Retain Files After Full Solve   | Yes                |
| Combine Restart Files           | Program Controlled |
| <b>Nonlinear Controls</b>       |                    |
| <b>Output Controls</b>          |                    |

Outline

- Project
  - Model (B4)
    - Geometry
    - Coordinate Systems
    - Mesh
    - Large Deflection-Linear Mat'l (B5)
      - Analysis Settings
      - Fixed Support
      - Force
    - Solution (B6)
      - Solution Information
      - Total Deformation
      - Equivalent Stress
      - Force Reaction

Details of "Force"

|                   |                          |
|-------------------|--------------------------|
| <b>Scope</b>      |                          |
| Scoping Method    | Geometry Selection       |
| Geometry          | 1 Edge                   |
| <b>Definition</b> |                          |
| Type              | Force                    |
| Define By         | Components               |
| Coordinate System | Global Coordinate System |
| X Component       | 350. N (ramped)          |
| Y Component       | 0. N (ramped)            |
| Z Component       | 0. N (ramped)            |
| Suppressed        | No                       |

B: Large Deflection-Linear Mat'l  
Force  
Time: 1. s  
11/18/2010 10:31 AM  
Force: 350. N  
Components: 350., 0., 0. N

- From Analysis Settings, change Auto Time Stepping definition and values and Restart Controls as shown.

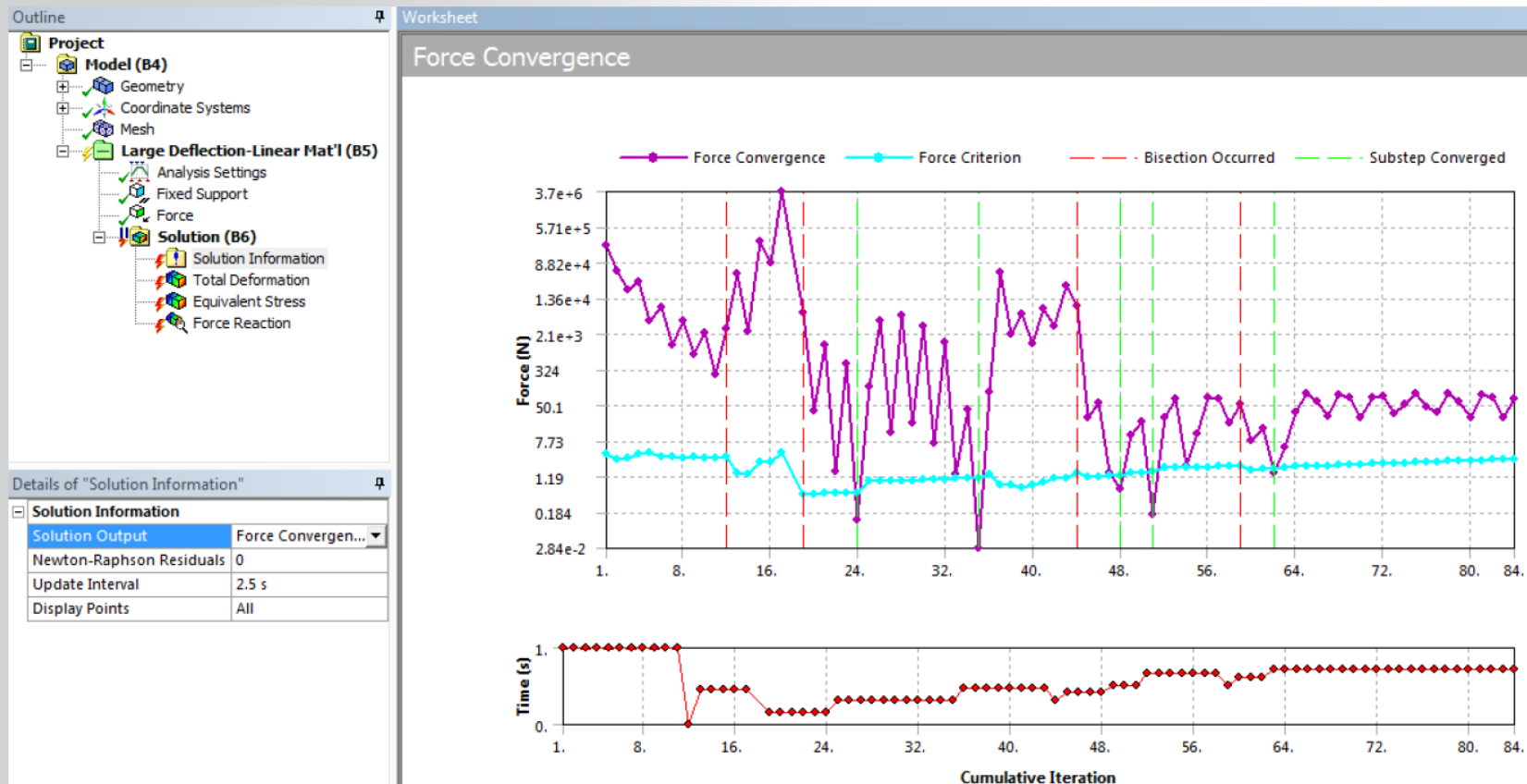
- Execute a new Solve...

## ... Workshop 2B: Performing a Restart

After a few warnings, errors and bisections, the run fails to converge.

- Review the Solution Information Solver Output and Force Convergence plot

```
*** ERROR ***                               CP =      44.850   TIME= 10:37:16
Solution not converged at time 0.715 (load step 1 substep 6) .
Run terminated.
```



## Examine the Solver Output text

- Note the warnings and errors recorded early in the run that indicate excessive element distortion and recommend more substeps as one possible remedy.

```
*** WARNING ***                      CP =      12.995    TIME= 09:49:01
There are 1033 small equation solver pivot terms.
    EQUIL ITER    6 COMPLETED.  NEW TRIANG MATRIX.  MAX DOF INC=  -8897.
```

```
*** ERROR ***                        CP =      13.088    TIME= 09:49:01
Excessive distortion of Element 1015.
```

```
*** ERROR ***                        CP =      13.088    TIME= 09:49:01
Excessive distortion of Element 1150.
```

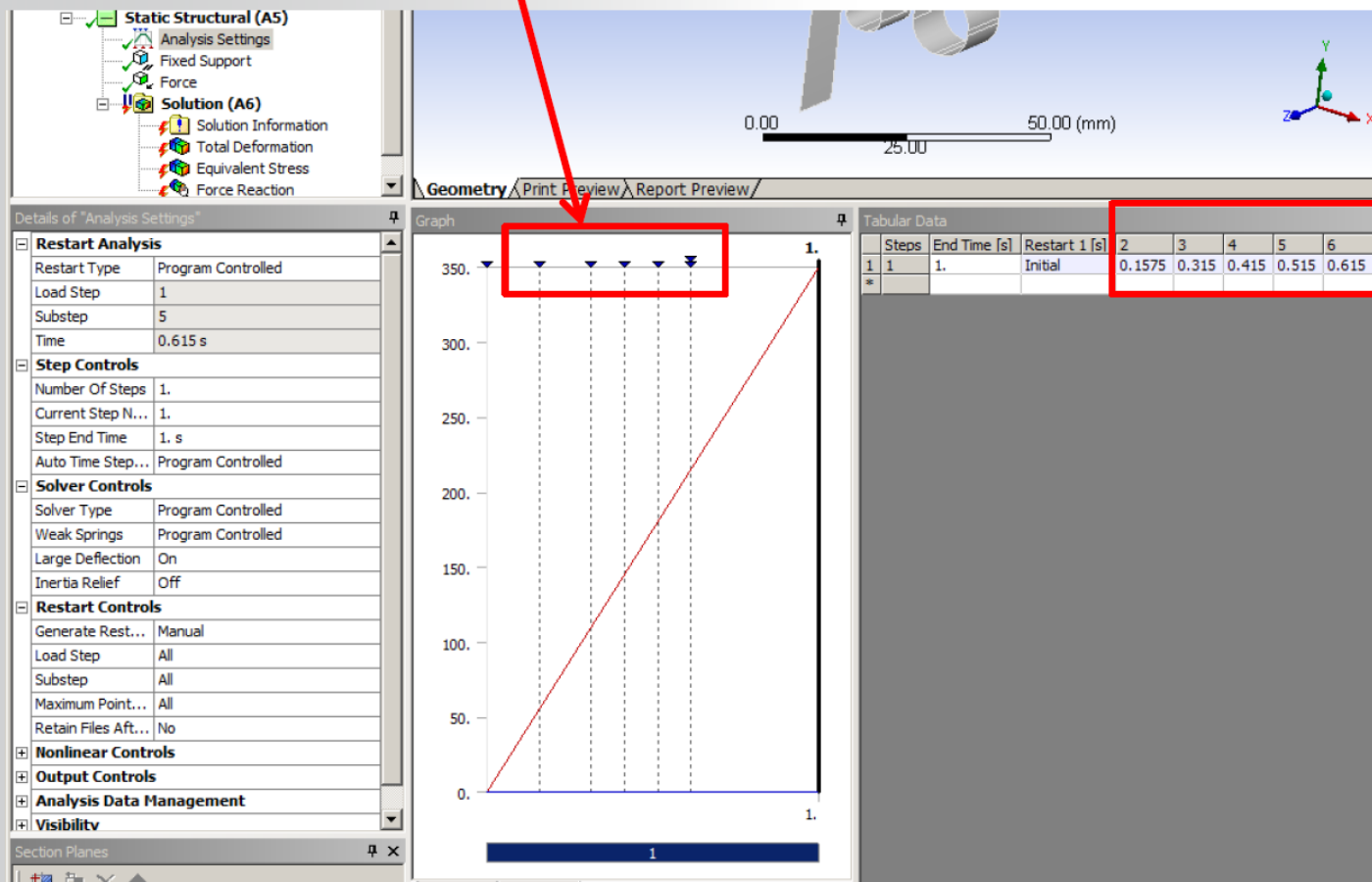
```
*** NOTE ***                         CP =      13.088    TIME= 09:49:01
One or more elements have become highly distorted.  Excessive
distortion of elements is usually a symptom indicating the need for
corrective action elsewhere.  Try incrementing the load more slowly
(increase the number of substeps or decrease the time step size).  You
may need to improve your mesh to obtain elements with better aspect
ratios.  Also consider the behavior of materials, contact pairs,
and/or constraint equations.  If this message appears in the first
iteration of first substep, be sure to perform element shape checking.
*** LOAD STEP      1    SUBSTEP      1 NOT COMPLETED.  CUM ITER =      19
*** BEGIN BISECTION NUMBER    2    NEW TIME INCREMENT=  0.15750
```



## ... Workshop 2B: Performing a Restart

Return to Analysis Settings,

- Highlight the Graphics Tab
- Note the total number (5) of “replayable” (blue) restart points matches the total number of converged substeps, as expected.



## ... Workshop 2B: Performing a Restart

Set Restart Type to “Manual”

- Set Current Restart Point to the last successfully converged subset (Loadstep 1, Substep 5)

Details of "Analysis Settings"

|                       |                        |
|-----------------------|------------------------|
| Restart Analysis      |                        |
| Restart Type          | Manual                 |
| Current Restart Point | Load Step 1, Substep 5 |
| Load Step             | 1                      |
| Substep               | 5                      |
| Time                  | 0.615 s                |

A logical first attempt to remedy this nonconvergence would be to simply enlarge the time step range from the default values.

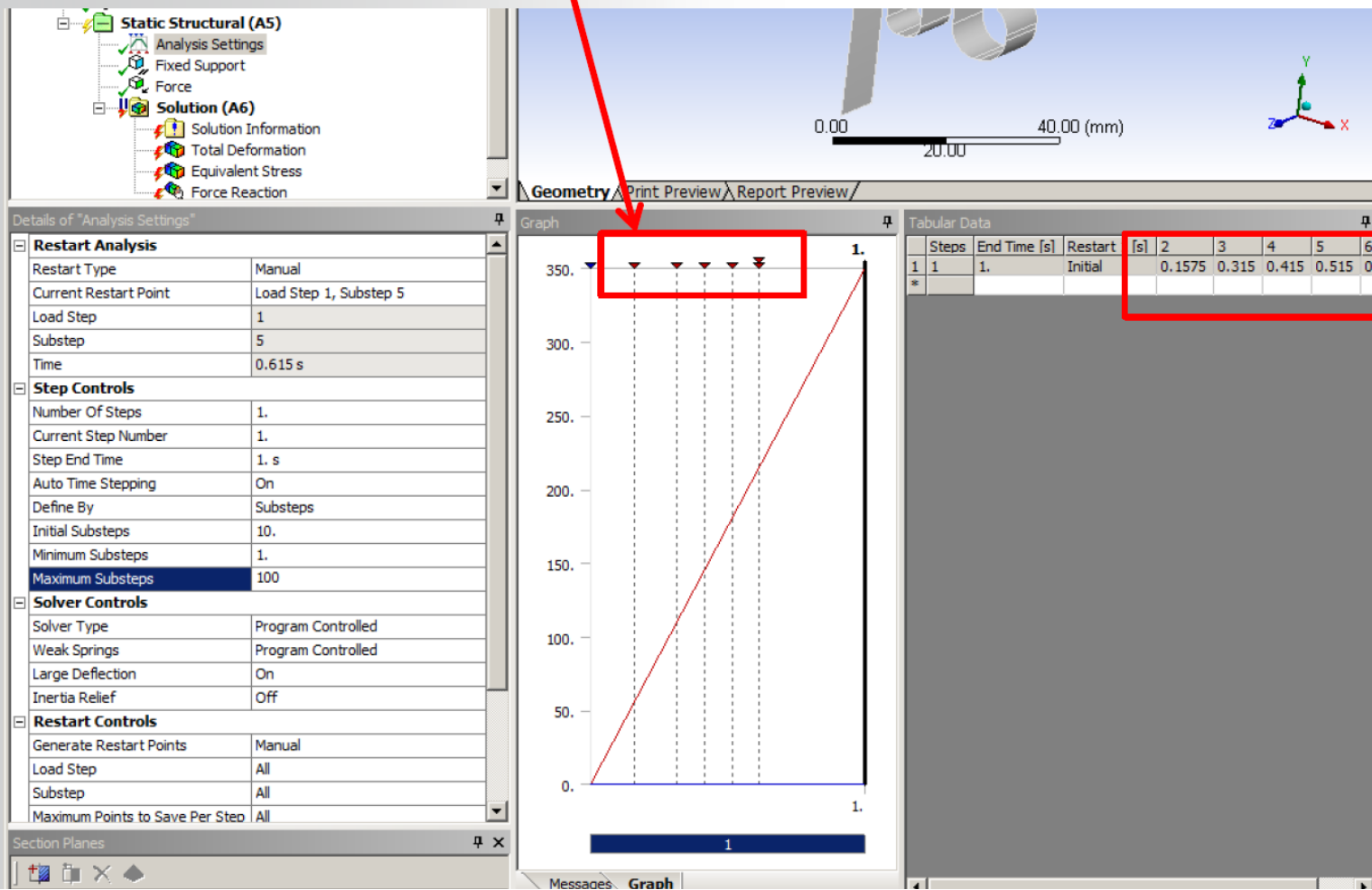
- Turn Auto Time Stepping ON and adjust the range as shown
  - 10 is a reasonable value to restart initially
  - 100 is just a guess based on experience.
- The range is highly problem dependent. Other models might require many more substeps.

Details of "Analysis Settings"

|                       |                        |
|-----------------------|------------------------|
| Restart Analysis      |                        |
| Restart Type          | Manual                 |
| Current Restart Point | Load Step 1, Substep 5 |
| Load Step             | 1                      |
| Substep               | 5                      |
| Time                  | 0.615 s                |
| Step Controls         |                        |
| Number Of Steps       | 1.                     |
| Current Step Number   | 1.                     |
| Step End Time         | 1. s                   |
| Auto Time Stepping    | On                     |
| Define By             | Substeps               |
| Initial Substeps      | 10.                    |
| Minimum Substeps      | 1.                     |
| Maximum Substeps      | 100                    |
| Solver Controls       |                        |
| Solver Type           | Program Controlled     |
| Weak Springs          | Program Controlled     |
| Large Deflection      | On                     |
| Inertia Relief        | Off                    |

# ... Workshop 2B: Performing a Restart

- Note the “replayable” restart points are now designated as “non-replayable”(red) because if the analysis were to be rerun from the very beginning with these new autotime stepping specifications, it would likely not produce the same substep solutions that are currently available for restarts.



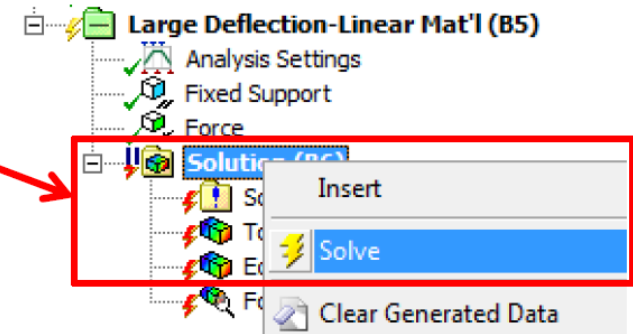


# ... Workshop 2B: Performing a Restart

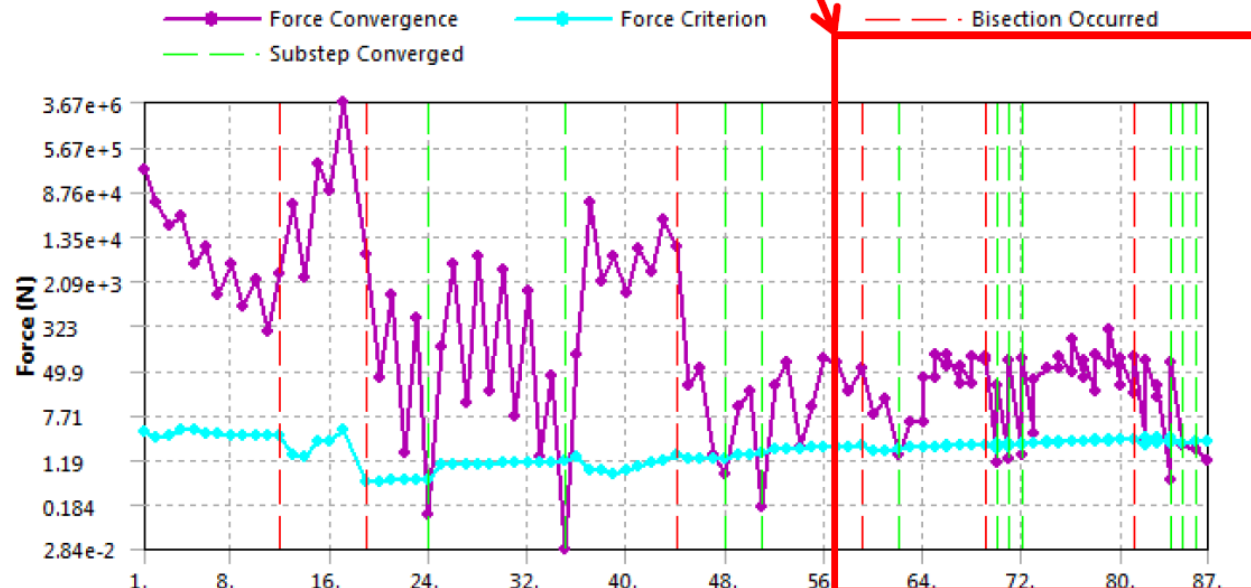
After making these changes, execute the restart.

Highlight the Solution information Branch and monitor the Force Convergence

- Notice the additional bisections that lead to a different convergence pattern and ultimately a successfully completed run



Force Convergence

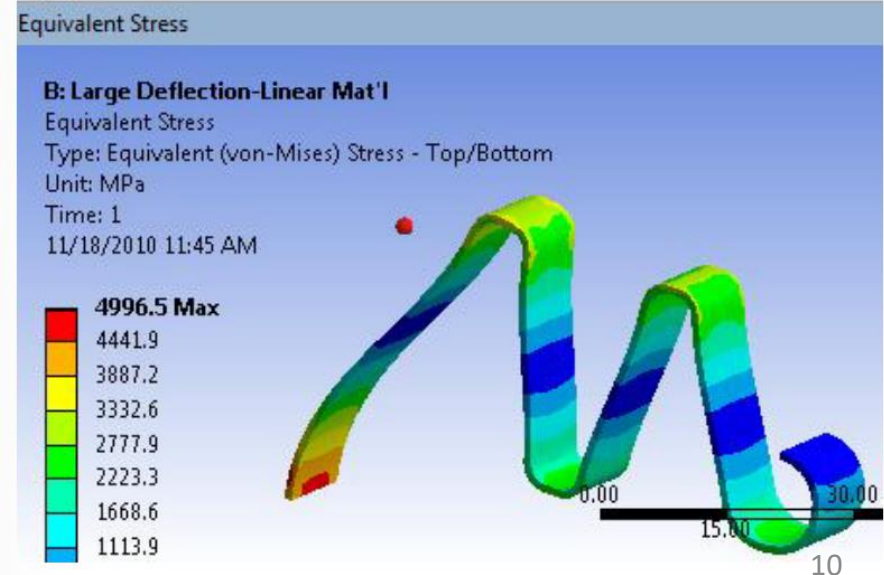
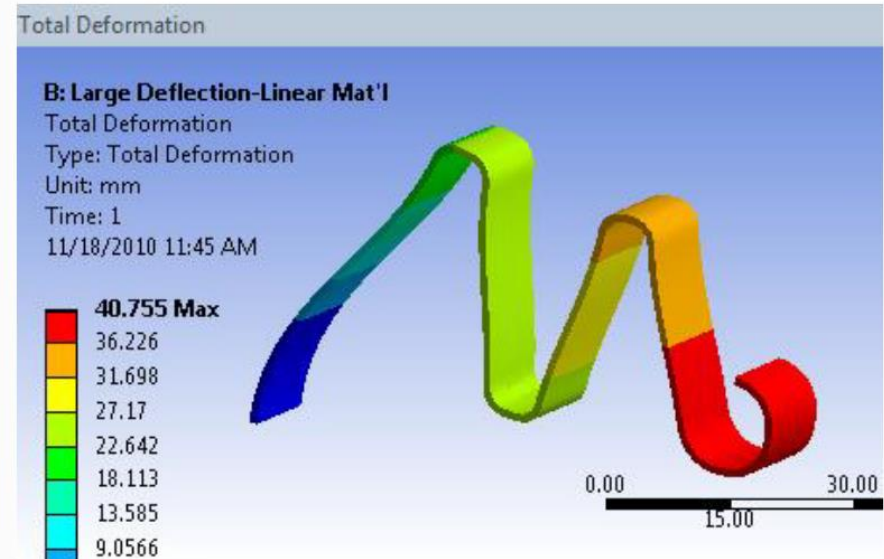


# ... Workshop 2B: Performing a Restart

Review the stress and displacement results as before.

Also confirm that the full load is reacted out at the fixed support

|                                 |                          |
|---------------------------------|--------------------------|
| Force Reaction                  |                          |
| Details of "Force Reaction"     |                          |
| [-] Definition                  |                          |
| Type                            | Force Reaction           |
| Location Method                 | Boundary Condition       |
| Boundary Condition              | Fixed Support            |
| Orientation                     | Global Coordinate System |
| [-] Options                     |                          |
| Result Selection                | All                      |
| Display Time                    | End Time                 |
| [+] Results                     |                          |
| [-] Maximum Value Over Time     |                          |
| <input type="checkbox"/> X Axis | -55.125 N                |
| <input type="checkbox"/> Y Axis | 1.5606e-003 N            |
| <input type="checkbox"/> Z Axis | 8.1334e-003 N            |
| <input type="checkbox"/> Total  | 350. N                   |
| [-] Minimum Value Over Time     |                          |
| <input type="checkbox"/> X Axis | -350. N                  |



## ... Workshop 2B: Performing a Restart

Having achieved convergence, we will now use restart technology to add an additional load step and define a new load without having to start the run from the beginning.

Return to Analysis Settings again

- Set Current Restart Point to the last successfully converged subset (Loadstep 1, Substep 12) →
- Set Number of Steps to "2" →
- With Current Step Number set to "2", → define the same Auto Time Stepping specifications for LS2 as were used for LS1

### Details of "Analysis Settings"

#### Restart Analysis

|                       |                         |
|-----------------------|-------------------------|
| Restart Type          | Manual                  |
| Current Restart Point | Load Step 1, Substep 12 |
| Load Step             | 1                       |
| Substep               | 12                      |
| Time                  | 1. s                    |

#### Step Controls

|                      |          |
|----------------------|----------|
| Number Of Steps      | 2.       |
| Current Step Number  | 2.       |
| Step End Time        | 2. s     |
| Auto Time Stepping   | On       |
| Define By            | Substeps |
| Carry Over Time Step | Off      |
| Initial Substeps     | 10.      |
| Minimum Substeps     | 1.       |
| Maximum Substeps     | 100.     |

#### Solver Controls

|                  |                    |
|------------------|--------------------|
| Solver Type      | Program Controlled |
| Weak Springs     | Program Controlled |
| Large Deflection | On                 |
| Inertia Relief   | Off                |

#### Restart Controls

# ... Workshop 2B: Performing a Restart

Highlight the existing "Force" in the Project Tree

- Define a new Y component of force (=100N) in the Tabular Data such that its application begins at LS 2

The screenshot displays the ANSYS Workbench interface for a Static Structural analysis. The Project Tree on the left shows the hierarchy: Static Structural (B5) > Analysis Settings > Fixed Support > Force. The Force is highlighted with a red circle. The main area shows the Force definition details and a graph of the force components over time.

**Details of "Force"**

| Scope          |                    |
|----------------|--------------------|
| Scoping Method | Geometry Selection |
| Geometry       | 1 Edge             |

| Definition                           |                          |
|--------------------------------------|--------------------------|
| Type                                 | Force                    |
| Define By                            | Components               |
| Coordinate System                    | Global Coordinate System |
| X Component                          | Tabular Data             |
| Y Component                          | Tabular Data             |
| <input type="checkbox"/> Z Component | 0. N (ramped)            |
| Suppressed                           | No                       |

**Graph**

The graph shows the force components over time. The X-axis represents time in seconds (0 to 2), and the Y-axis represents force in Newtons (0 to 350). The X component (red line) ramps up from 0 to 350 N at 1 second and remains constant. The Y component (green line) is 0 N until 1 second, then ramps up to 100 N at 2 seconds. The Z component (blue line) is 0 N throughout.

**Tabular Data**

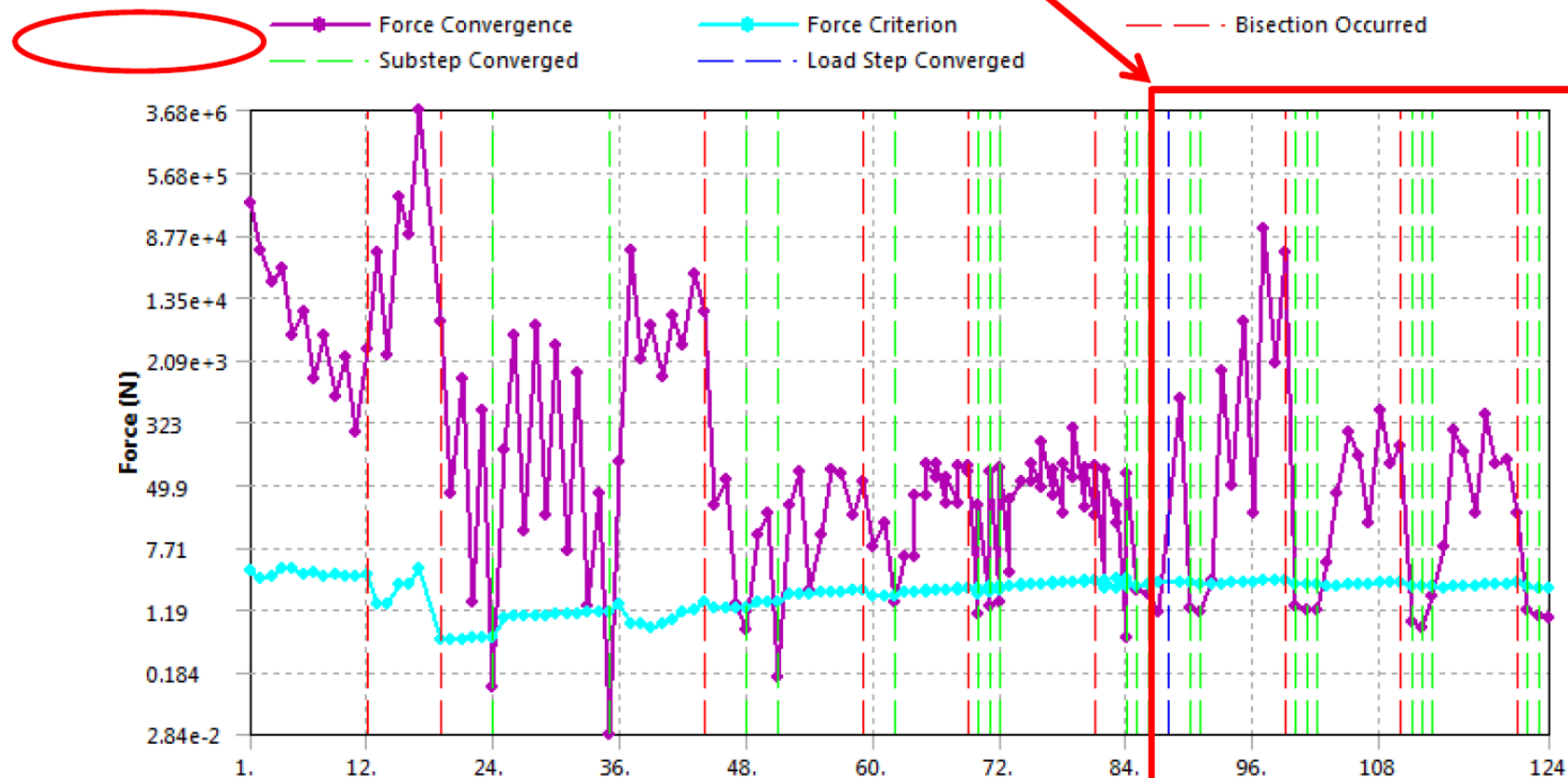
| Steps | Time [s] | <input checked="" type="checkbox"/> X [N] | <input checked="" type="checkbox"/> Y [N] | <input checked="" type="checkbox"/> Z [N] |
|-------|----------|-------------------------------------------|-------------------------------------------|-------------------------------------------|
| 1 1   | 0.       | 0.                                        | 0.                                        | 0.                                        |
| 2 1   | 1.       | 350.                                      | 0.                                        | 0.                                        |
| 3 2   | 2.       | = 350.                                    | 100.                                      | = 0.                                      |
| *     |          |                                           |                                           |                                           |

## ... Workshop 2B: Performing a Restart

After making these changes, re execute the solve.

- Note, from the force convergence graph, that the run picks up at the end of LS1, as expected, and runs successfully to completion.

Force Convergence

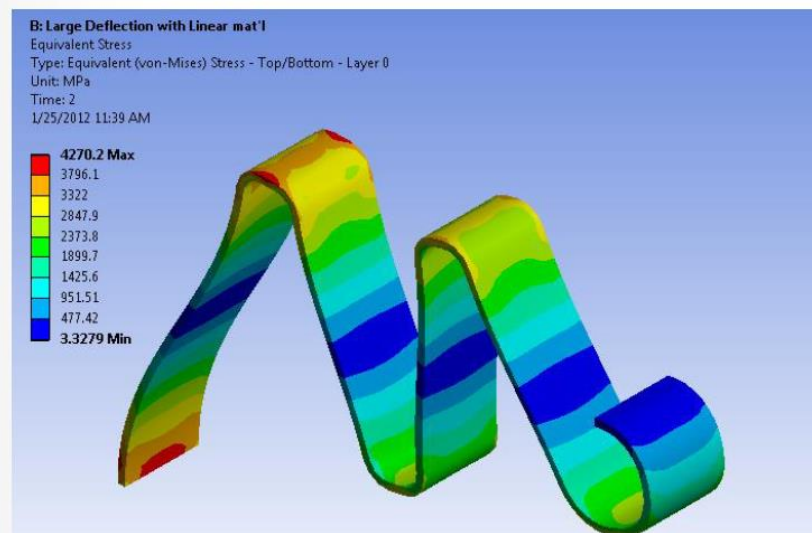
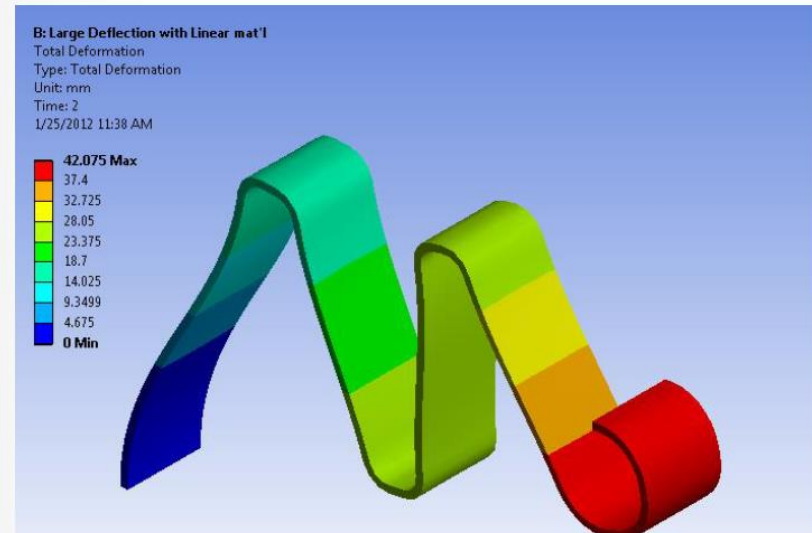




# ... Workshop 2B: Performing a Restart

Review the stress and displacement results and reaction forces before.

| Details of "Force Reaction"     |                          |  |
|---------------------------------|--------------------------|--|
| Location Method                 | Boundary Condition       |  |
| Boundary Condition              | Fixed Support            |  |
| Orientation                     | Global Coordinate System |  |
| Suppressed                      | No                       |  |
| <b>Options</b>                  |                          |  |
| Result Selection                | All                      |  |
| Display Time                    | End Time                 |  |
| <b>Results</b>                  |                          |  |
| <b>Maximum Value Over Time</b>  |                          |  |
| <input type="checkbox"/> X Axis | -55.125 N                |  |
| <input type="checkbox"/> Y Axis | 1.5606e-003 N            |  |
| <input type="checkbox"/> Z Axis | 8.1333e-003 N            |  |
| <input type="checkbox"/> Total  | 364.01 N                 |  |
| <b>Minimum Value Over Time</b>  |                          |  |
| <input type="checkbox"/> X Axis | -350. N                  |  |
| <input type="checkbox"/> Y Axis | -100. N                  |  |
| <input type="checkbox"/> Z Axis | -2.4197e-002 N           |  |
| <input type="checkbox"/> Total  | 55.125 N                 |  |
| <b>Information</b>              |                          |  |





# ... Workshop 2B: Performing a Restart

For comparison, clear the generated data and re-solve the model from the beginning.

- Notice the model now solves with only three bisections over 2 load steps and in 74 total iterations (as compared to 124 iterations), cutting the run time significantly. This underscores the importance of taking advantage of auto time stepping technology with a sufficient range in nonlinear problems.

